Static

```
Public class Rectangle
{
  private int length;      <- set / get
  private int width;       <- set / get
  private static int noofobjects;  -> get

  Public Rectangle( )
  {
    length = 100;
    width = 300;
    noofobjects ++;
  }
  Public void setlength (int L)
  {
    length = L;
  }
  Public int getlength ( )
  {
    return length;
  }
  Public Static int getnoofobjects( )
  {
    return noofobjects;
  }
}
```

```
Public class test
{
  Public static void main(String[]args)
  {
    Rectangle r1 = new Rectangle();
    Rectangle r2 = new Rectangle();
    Rectangle r3 = new Rectangle();
    Sy — (r3.getnoofobjects());
    Sy — (r1.getnoofobjects());
    Sy — (r2.getnoofobjects());

    Sy — (Rectangle.getnoofobjects())
  }
}
```

Math.sqrt(----)

Math → class

sqrt → method

Static

Math m = new Math().

M.sqrt(

Memory

Rectangle (r1)

length: 100

Width: 300

noofobject: ~~1~~ ~~2~~ 3

Rectangle (r2)

length: 100

Width: 300

noofobject: ~~2~~ 3

Rectangle (r3)

length: 100

Width: 300

Noofobjects: 3

3
3
3
3

n~

```
in(String[]args)
w Rectangle();
Rectangle();
Rectangle();
noofobjects());
noofobjects());
noofobjects());

e.getnoofobjects()

(----)

ethod
```

Utility class

```
Public class Calc
{
Public Static int sum(int x, int y)
{ return x+y;
}
Public Static int Mul(int x, int y)
{ return x*y;
}
```

```
Public class test
{
Public static void main(String[]args)
{ Calc   = new calc();
Sy ___ (c.sum(5,4));
Sy ___ (Calc.sum(5,4));
}
}
```

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```
Public class test
{ Public static void main(String[]args)
{ display();
}
Public Static Void display()
{ Sy ___ ("Hello");
}
```

```
test.display();
test t=new test();
t.display();
```

```
Math m= new Math();
M.sqrt(-.)
```

this

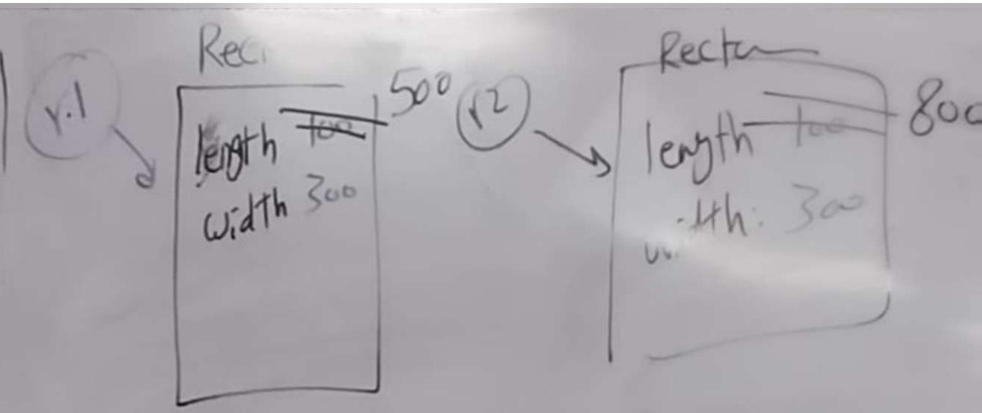
```
Public class Rectangle
{
    private int length;     set get
    private int width;      set get
    private static int no.ofobjects;

    Public Rectangle( )
    { length=100
      width=300
    }

    Public void setlength(int Length)
    { this.Length = Length;
    }
    Public int getlength( )
    { return length;
    }
    Public Static int getnoofobjects( )
    { return noofobjects;
    }
}
```

refers to current object

```
Public class test
{
Public Static Void main(String[] args)
{
Rectangle r1= new Rectangle();
Rectangle r2= new Rectangle();
r1.setLength(500);
r2.setLength(800);
```

Length = L;

this.length=length;



this

Public class Rectangle
{
private int length; → instance variable
private int width;
private static int no.ofobjects; → class variable

Datafields / attributes / members — Data

Methods

Public Rectangle( )
{ length=100
width=300
}

Public void setlength (int Length)
{ this.Length = Length;
}
refers to current object

Public int getlength ( )
{ return length;
this.length;
}

Public Static int getnoofobjects( )
{ return noofobjects;
}

this.

Length = L;
this.length=length;

Public int display (int x)
{ int y=3*x;
return y;
}

local variable

Parameter or argument

Types of variables in OOP

A a = new A();

Data
methods

members (attributes)
→ instance variable
→ class " (Static)

local variable (inside the method)

Parameters (arguments)

methods

Ex Public class A
{ Private int X=10;
Public void m ( )
{ int X=5;
sy — ("X="+X);
}

instance var

local variable hides the instance variable

X=5

public class A
{ Private int X=10;
Public void m ( )
{ int X=5;
sy — ("X="+ this.X);
}
}

X=10

Public Class test
{ Public Static void main( )
{ A a=new A( );
a.m( );
}
}